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AI implementation of Computer Vision into Medical Imaging

Analysis:

Computer vision is the application of having AI programmed, trained, and implemented for the purpose of interpreting and analyzing visual information and media such as pictures and videos. Since its inception, computer vision has been heavily implemented into medical imaging with the objective of better enhancing, increasing efficiency and further progressing analyzation in a wide range of medical practices. As with anything, drawbacks should always be considered. With AI's current capabilities and future potential, implementation, both present and future should be held up to scrutiny it faces.

Computer Visioning analyzes medical imaging by using AI algorithms to analyze thousands of photos of previous medical imaging containing normal screenings and screenings with anomalies. Once recognized, CNNs then use pattern recognition and other features to learn what to look for, and combining with AI's ability to rapidly learn, this results in AI, with efficient training, the ability to accurately predict and analyze imaging and recognition of certain signs of a particular diagnosis, drawing back to conclusions to accurately make diagnosis, location, and exact shape. In addition, CNNs have the ability to enhance medical imaging resolution themselves, further improving on diagnosis process by recognizing perhaps unidentified anomalies. This has resulted in a rapid push to

introduce CNNs to several areas of R&D within medical imaging, and potentially into more human applications of medical field, such as surgery.

Reflection:

The current landscape of AI in medical imaging has been new, and because of rapid development, it has been in turn, rapidly implemented in several fields across medical imaging. Substantial improvements have been seen in making fields such as X-rays, CT scans, and other scans, while making remarkable improvements. The potential for this could be monumental in rapidly not only producing quality of medicine, but leading to better satisfaction with all parties involved, whether that be patient remission and lack of burden on patient, advancement of fields of study, and newly discoverable ways to overall improve quality of life, while becoming more environmentally sound and reducing resource allocation.

For these reasons not only to be valued by the medical field, but the advocates and detractors alike, Computer Vision in medical imagery must hold up to scrutiny as well. The most obvious drawback that has been cited by detractors is the current new inception of implementation, and rapid adoption, creating a sense of uneasiness and perceived instability. In addition to ethical concerns and privacy. In addition to a certain scenario that has been seen with rapid adoption of AI that must especially not be is the increasing and concerning trend of that could be tied into this scenario is Automation Bias, or the tendency to blindly trust answers computer input over human thought, which can lead to a

potentially disastrous scenario of Cognitive Offloading, the process of reducing mental effort such as critical thinking, problem solving, pattern recognition, and Bias recognition on to external tools.

Nevertheless, the potential for computer vision in medical imagery could be monumental and revolutionary in medicine and quality of life, but its detachments must be held up to the same standards to achieve these standards and potential goals at the maximum. Although solutions will not be perfect, they still must be strived for, with those in the medical field and Computer Vision as a synergetic relationship instead of potential dependency on one another.